Giacomo Negrinelli

About

Senior Civil and Hydraulic Engineer (Ing., MSc, BSc) and Project Manager with over 11 years of international experience in hydropower, dam safety, water resources, river engineering, and flood risk management. My expertise spans research, design, and construction, developed at leading organisations including Deltares (Netherlands), Frosio Next (Italy), Mott MacDonald (UK), and Gruner Stucky (Switzerland/Georgia). I am professionally qualified as Engineer in Italy (State Examination at University of Padua).

Work Experience

Project Manager / Engineer
Gruner Stucky SA, Renens (Switzerland)
Aug 2021 – Present

Within the Energy Business Area, I provide engineering consultancy services for the design and construction of dams and hydropower plants.

- Project management and design coordination.
- Hydropower planning and design.
- Design of hydraulic structures and hydraulic modelling.
- Dam safety and emergency preparedness.
- Geospatial spatial analysis.

Project Engineer
Gruner Stucky Caucasus Ltd, Tbilisi (Georgia)
Oct 2021 – Dec 2023 · 2 yrs 3 mos

Office-based secondment to Gruner Stucky Caucasus Ltd, part of Gruner AG.

Civil Engineer
Mott MacDonald Ltd, Cambridge (United Kingdom)
Jan 2018 – Jul 2021 · 3 yrs 7 mos

Within the Dams, Rivers, and Geotechnics Group, I provided engineering consultancy services for the design and construction of dams, reservoirs and river hydraulic structures.

- Design and construction supervision of hydraulic structures.
- 1D/2D and CFD modelling of hydraulic structures.
- Project management and administration.
- Geospatial spatial analysis.

Site Engineer
JN Bentley Ltd, Great Yarmouth (United Kingdom)
Oct 2019 – Mar 2020 · 6 mos

Site-based secondment for the Environment Agency Great Yarmouth Flood Defences Epoch 2 project.

- Setting-out, levelling and surveying.
- Verification of plans and drawings, and coordination of construction.
- Quality control and implementation of health and safety policies on site.

Hydraulic Engineer
Frosio Next S.r.l., Brescia (Italy)
Jan 2016 – Dec 2017 · 2 yrs

Within the BKW Engineering Network, I provided engineering consultancy services for the design, construction, and operation of hydropower schemes.

- Due diligence of hydropower plants.
- Hydrological analysis and hydropower potential assessment.
- 1D/2D hydraulic modelling.
- Geospatial spatial analysis.

Language Proficiency

- **Italian** | *Native*
- **English** | *Full professional*
- **French** | *Limited working*

Countries of Work Experience

Australia · Belgium · Burkina Faso · Chile · Georgia · Italy · Laos · Mozambique · Philippines · Serbia · Slovenia · Switzerland · Tajikistan · Turkmenistan · UK · USA · Uzbekistan · Zambia · Zimbabwe.

Certifications

- **University of Padua**
  Section A Civil and Environmental Engineering Italian State Exam.
- **UK Construction Industry Training Board (CITB)**
  Site Management Safety Training Scheme (SMSTS), CSCS Academically Qualified Person Card.
- **BuildingSMART International**
  BIM Professional Certification Foundation.

Training

- Communication Pour Chefs de Projet at ISMAT Group SA, Lausanne (2026).
- Dam Risk Analysis Training at GNCOLD, Tbilisi (2023).
- BIM Methodology Training at Gruner AG, Renens (2023).
- 4x4 Off Road Training Course at Explore 4×4, Thetford (2021).
- Building Client Relationships and Collaboration at Mott MacDonald Academy, London (2019).
- FLOW-3D Workshop at EDF, Lyon (2019).
- Project Management Essentials at Mott MacDonald Academy, Croydon (2018).
- Presentation Skills at Mott MacDonald Academy, Cambridge (2018).

Teaching Assistant
University of Brescia, Brescia (Italy)
Feb 2017 – Sep 2017 · 8 mos

Teaching assistant for the course "Hydrology" (702998), Bachelor's Degree in Environmental and Land Engineering.

Hydraulic Engineer
Dizeta Ingegneria Studio Associato, Milano (Italy)
Nov 2015 – Dec 2015 · 2 mos

Freelance collaboration on a feasibility study for the reconstruction of a concrete dam on the Sauro River for water supply and irrigation, in the municipality of Agro di Aliano (Italy).

Graduate Intern
Deltares, Delft (Netherlands)
Mar 2015 – Aug 2015 · 6 mos

Research within the Dike Safety Department into flood embankment failure caused by backward erosion piping, involving physical and numerical modelling of seepage and erosion.

Graduate Intern
Consorzio di Bonifica Oglio Mella, Flero (Italy)
Jul 2014 – Sep 2014 · 3 mos

Internship with the public authority responsible for water resources management in the Oglio and Mella river basins.

- Streamflow measurement campaign.
- Technical assessment of hydropower projects.
- Hydraulic modelling of irrigation canals.

Education

Master's degree, Land and Environmental Engineering
University of Brescia, Brescia (Italy)
2013 –2015

- **Grade:** 110/110 cum laude.
- **Main subjects:** Structural Design, Geotechnics, Hydraulic Structures, Environmental Hydraulics, Water and Wastewater Treatment, Geomatics, Urban planning, Risk Mitigation.
- **Thesis:** Investigation of the process of seepage and backward erosion piping under dikes in heterogeneous sands.

Erasmus+ Exchange Program, Civil Engineering
Technical University of Munich, München (Germany)
2014 –2015

- **Grade:** 1.4 (Excellent).
- **Main subjects:** River Engineering, Hydro-morphology, Water Management, Alpine Hazards, Land Management and Policy, GIS.

Bachelor's degree, Civil Engineering
University of Brescia, Brescia (Italy)
2009 –2013

- **Grade:** 110/110 cum laude.
- **Main subjects:** Mathematics, Physics, Chemistry, Structural Mechanics, Hydraulics, Hydrology, Geology, Surveying, Architecture, Building Technology, Economics.
- **Thesis:** An estimation procedure of synthetic design hydrographs for the design of flood control structures.

High School Diploma, Scientific Studies
Liceo Scientifico Statale "Nicolò Copernico", Brescia (Italy)
2004 –2009

- **Grade:** 100/100.
- **Main subjects:** Scientific (Mathematics, Physics, Chemistry, Biology, Earth Science) and humanistic (Italian, Latin, History, Philosophy) studies.
- **Education level:** Lyceum (five-year university-preparatory school).

Resume

- HEC-RAS 2D Modelling Workshop at JBA Consulting, London (2018).
- Project Management course at ProjectGroup S.r.l., Brescia (2017).
- HEC-RAS 2D training course at H.S. S.r.l., Capannori (2016).
- Work Health and Safety training course at ConsulGroup S.r.l., Brescia (2016).

Publications

- Aykut, M., Micheloud, S., **Negrinelli, G.**, Bieri, M.P., Galdava, L., & Kopadze, T. (2024). *The Mtkvari hydropower project in Georgia: Dam safety upgrade of a project under execution*. In 2024 International Conference and Exhibition on Hydropower and Dams, Graz (Austria).
- Micheloud, S., **Negrinelli, G.**, Aykut, M., Galdava, L., & Kopadze, T. (2022). *A modern design approach at Mtkvari, Georgia*. Hydropower & Dams, 29(6).
- Cazzago, D., Papetti, L.L., & **Negrinelli, G.** (2016). *Design optimization of a very low head power plant - plenty of hydraulic common sense and a bit of CFD*. In 2016 International Conference and Exhibition on Hydropower and Dams, Montreux (Switzerland).
- De Lange, D.A., Van Beek, V.M., Schenkeveld, F.M., Van der Linden, T.I., & **Negrinelli, G.** (2016). *Preparation techniques for unconventional sand samples*. In 3rd European Conference on Physical Modelling in Geotechnics, Nantes (France).
- **Negrinelli, G.**, Van Beek, V.M., & Ranzi, R. (2016). *Experimental and numerical investigation of backward erosion piping in heterogeneous sands*. In 8th International Conference on Scour and Erosion, Oxford (UK).
- **Negrinelli, G.**, Van Beek, V.M., & Ranzi, R. (2016). *Small-scale piping experiments in heterogeneous sands*. In XXXV National Conference on Hydraulics and Hydraulic Structures, Bologna (Italy).
- Van Beek, V.M., **Negrinelli, G.**, Koelewijn, A.R., & Förster, U. (2015). *A coarse sand barrier as an effective piping measure*. Geotechniek, September 2015, 4-7.